

SIMULATION OF VOLT-AMPERE CHARACTERISTICS

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Volt-ampere (V-I) characteristics describe the behaviour of electronic components. These are simply the plots of applied voltage versus current flowing through a component or device. In most cases, V-I curves are the basis for selection of a particular component or device in order to meet the design parameters.

Here is a simple program that simulates V-I characteristics of resistors and diodes. In simulation, the empirical voltage-current relationships are utilised. It can be used to plot V-I characteristics of a component by changing V-I relationships in the program. This simulation program is designed and tested in Visual Basic 6 and MS Access under Windows XP environment.

As per Ohm's law ($V=IR$), the empirical voltage-current relationship of a resistor is:

$$I = V/R$$

The empirical current-voltage relationship of a diode is expressed as:

$$I = I_0 (e^{(V/\eta V_T)} - 1)$$

where $\eta=1$ for germanium diode and $\eta=2$ for silicon diode, $V_T = 26$ mV at 300°C and I_0 (reverse saturation current) $= 50 \times 10^{-9}$ amperes. Using these values, you can plot V-I characteristics of either a germanium diode or silicon diode.

The V-I plots for a few steps of initial voltages of a 100-ohm resistor and a germanium diode are shown in Screenshots 1 and 2, respectively. In either case, you can change the voltage steps at your convenience. Soon after by pressing 'Tab' or 'Enter' key, the current is calculated automatically in the program and the plot changes accordingly. Alternatively, you can get each of these plots by selecting the corresponding option buttons at run time.

You can change the resistance



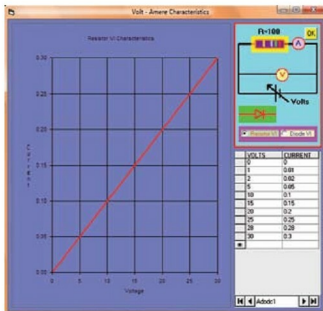
value at run time. The V-I output can be observed after clicking 'Ok' button. The V-I plot of a silicon diode can be obtained by changing the value of η to '2' in the program.

Form design

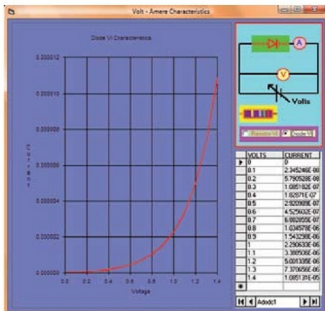
Install Visual Basic in your PC and launch the application from the desktop. Open 'new project' to get a new form. On the form, place the required controls and resize them as shown in Screenshot 3. Also, set their properties as listed in the table.

MS Access database

Open MS Access application and create database tables for the resistor and diode. Save the tables as 'ohmslaw' and 'diodev', respectively. Create two fields for each table with field names as 'volts' and 'current,' and field types



Screenshot 1: V-I characteristics of a resistor



Screenshot 2: V-I characteristics of a diode